

SHETH L.U.J. & SIR M.V. COLLEGE OF SCIENCE

SUBJECT - Data Analysis with SAS / SPSS / R

Aim :- Reshaping data using pivot_longer()/pivot_wider() (R).

Output :-

The first screenshot shows the initial data preparation and the first pivot operation. The console output is as follows:

```
> library(tidyverse)
> # 1. SETUP: Prepare Data
> df <- mtcars %>%
+   mutate(CarID = row_number(),
+          Car = rownames(mtcars)) %>%
+   select(CarID, Car, cyl, mpg, hp)
> print("1. Original wide data")
[1] "1. Original wide data"
> print(head(df))
```

	CarID	Car	cyl	mpg	hp
Mazda RX4	1	Mazda RX4	6	21.0	110
Mazda RX4 Wag	2	Mazda RX4 Wag	6	21.0	110
Datsun 710	3	Datsun 710	4	22.8	93
Hornet 4 Drive	4	Hornet 4 Drive	6	21.4	110
Hornet Sportabout	5	Hornet Sportabout	8	18.7	175
Valiant	6	Valiant	6	18.1	105

The second screenshot shows the second pivot operation. The console output is as follows:

```
> # 2. PIVOT_LONGER (Wide to Long)
> # Combine mpg and hp into a single column for comparison
> long_df <- df %>%
+   pivot_longer(
+     cols = c(mpg, hp),
+     names_to = "Metric",
+     values_to = "Value"
+   )
> print("2. Long Format (pivot_longer)")
[1] "2. Long Format (pivot_longer)"
> print(head(long_df, 6))
```

	CarID	Car	cyl	Metric	Value
1	1	Mazda RX4	6	mpg	21
2	1	Mazda RX4	6	hp	110
3	2	Mazda RX4 Wag	6	mpg	21
4	2	Mazda RX4 Wag	6	hp	110
5	3	Datsun 710	4	mpg	22.8
6	3	Datsun 710	4	hp	93

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The screenshot shows the RStudio interface with a script editor, console, and environment pane. The script defines a function to convert long data to wide format using `pivot_wider`. The console output shows the execution of the script, resulting in a wide format table with 6 rows and 5 columns.

```
R - R4.5.2 - / -  
> # Convert the long data back to wide format  
> wide_df <- long_df %>%  
+   pivot_wider(  
+     names_from = Metric,  
+     values_from = Value  
+   )  
> print("3. Wide Format (Back to Wide)")  
[1] "3. Wide Format (Back to Wide)"  
> print(head(wide_df))  
# A tibble: 6 x 5  
  CarID Car      cyl  mpg   hp  
  <int> <chr>    <dbl> <dbl> <dbl>  
1     1 Mazda RX4      6    21   110  
2     2 Mazda RX4 Wag    6    21   110  
3     3 Datsun 710      4   22.8    93  
4     4 Hornet 4 Drive    6   21.4   110  
5     5 Hornet Sportabout  8   18.7   175  
6     6 Valiant        6   18.1   105  
> # 4. ADVANCED EXAMPLE: Reshaping for Reporting  
> # Create a table where each cylinder count becomes a column,  
> # and values represent mpg  
> cyl_pivot <- df %>%  
+   select(CarID, cyl, mpg) %>%  
+   pivot_wider(  
+     names_from = cyl,  
+     values_from = mpg,  
+     names_prefix = "cyl_"  
+   )  
> print("4. Cylinder-wise MPG Table")  
[1] "4. Cylinder-wise MPG Table"  
> print(head(cyl_pivot))  
# A tibble: 6 x 4  
  CarID cyl_6 cyl_4 cyl_8  
  <int> <dbl> <dbl> <dbl>  
1     1    21    NA    NA  
2     2    21    NA    NA  
3     3    NA   22.8    NA  
4     4   21.4    NA    NA  
5     5    NA    NA   18.7  
6     6   18.1    NA    NA
```

The screenshot shows the RStudio interface with a script editor, console, and environment pane. The script defines a function to convert wide data to long format using `pivot_longer`. The console output shows the execution of the script, resulting in a long format table with 6 rows and 5 columns.

```
R - R4.5.2 - / -  
> # Create a table where each cylinder count becomes a column,  
> # and values represent mpg  
> cyl_pivot <- df %>%  
+   select(CarID, cyl, mpg) %>%  
+   pivot_wider(  
+     names_from = cyl,  
+     values_from = mpg,  
+     names_prefix = "cyl_"  
+   )  
> print("4. Cylinder-wise MPG Table")  
[1] "4. Cylinder-wise MPG Table"  
> print(head(cyl_pivot))  
# A tibble: 6 x 4  
  CarID cyl_6 cyl_4 cyl_8  
  <int> <dbl> <dbl> <dbl>  
1     1    21    NA    NA  
2     2    21    NA    NA  
3     3    NA   22.8    NA  
4     4   21.4    NA    NA  
5     5    NA    NA   18.7  
6     6   18.1    NA    NA
```

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